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Jeff Shen

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Education

Ph.D., Princeton University. Astrophysical Sciences. 2022 - Present
H.B.Sc., University of Toronto. Astronomy & Astrophysics and Statistics. 2018-2022

Publications

5. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, et. al. *Stellar parameters for 220m stars from Gaia DR3 BP/RP spectra*, in prep.

4. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, J. Ted Mackereth, Yuan-Sen Ting, Jo Bovy. *Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution*, in prep.

3. Seery Chen, Deborah M. Lokhorst, **Jeff Shen**, Imad Pasha, Evgeni I. Malakhov, Roberto Abraham, Pieter van Dokkum. *The Dragonfly Spectral Line Mapper: design and first light.* 2022, Proc. SPIE 12182, Ground-based and Airborne Telescopes IX, 121824E. arXiv: 2209.07489.

2. Deborah M. Lokhorst, Seery Chen, Imad Pasha, **Jeff Shen**, Evgeni I. Malakhov, Roberto G. Abraham, Pieter van Dokkum. *The pathfinder Dragonfly Spectral Line Mapper: pushing the limits for ultra-low surface brightness spectroscopy.* 2022, Proc. SPIE 12182, Ground-based and Airborne Telescopes IX, 121821T. arXiv: 2209.07487

1. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. *The Mass of the Milky Way from the H3 Survey.* 2022, ApJ, 925, 1. arXiv: 2111.09327.

0. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. *Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$.* 2021, ApJ, 917, 79. arXiv: 2105.11572.

Research Experience

2022 Jun - Present	**Estimating stellar parameters using Gaia DR3 BP/RP spectra** Supervisors: Dr. Joshua Speagle, Dr. Neige Frankel Independent research. Creating a catalog of high-quality stellar parameters from low resolution BP/RP spectra from DR3.
2021 May - Present	**Predicting the ages of stars with machine learning** Supervisors: Dr. Joshua Speagle, Dr. Neige Frankel, Dr. J. Ted Mackereth Independent research. Probabilistic predictions of stellar ages using normalizing flows. Quantifying the contribution of stellar/Galactic information to age estimates.
2021 Sep - 2022 Jun	**Narrowband upgrade for the Dragonfly Telephoto Array** Supervisor: Prof. Roberto Abraham Senior thesis. Containerized camera control software in C++ and Rust with real-time, multi-threaded data processing.
2021 Jul - 2022 Jun	**Constraining the strength of thermal tides with Stan** Supervisor: Prof. Norman Murray Independent research. Fitting dynamical models for the angular momentum of the Earth-Moon system to constrain the strength of thermal tides over geological time.

2020 Aug - 2021 Nov	**Estimating the mass of the Galaxy with Bayesian hierarchical models** Supervisors: Prof. Gwendolyn Eadie, Prof. Norman Murray, Dennis Zaritsky, Dr. Joshua Speagle Undergraduate fellowship with the Canadian Institute for Theoretical Astrophysics (CITA). Developed a hierarchical Bayesian model for modelling the mass of the Galaxy using a fast and scalable sampling algorithm. Collaborated with the H3 Survey team and presented results in ApJ.
2020 May - 2021 May	**Molecular gas in high-redshift galaxies with ALMA** Supervisor: Prof. Allison Man Summer undergraduate research program with the Dunlap Institute. Analyzed ALMA data to infer molecular gas masses in several gravitationally lensed high-redshift galaxies. Presented results in a first-author publication in ApJ.

Honours and Awards

2022 Jan	**Astrostatistics Student Paper Competition Finalist** Astrostatistics Interest Group of the American Statistical Association
2021 - 2022	**Dean's List Scholar** University of Toronto
2021 May - 2021 Aug	**Undergraduate Summer Research Fellowship, ($8,396 CAD)** Canadian Institute for Theoretical Astrophysics, University of Toronto
2020 May - 2020 Aug	**Summer Undergraduate Research Program Award ($9,500 CAD)** Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Talks

2022 August	**JSM 2022, Washington, D.C.** Contributed talk: "The Mass of the Milky Way from the H3 Survey."
2021 August	**SDSS 2021 Collaboration Meeting, JHU/Online** Lightning talk: "Predicting the ages of stars with machine learning."
2021 May	**Stellar Stats Workshop, UofT/Online** Lightning talk: "Estimating the mass distribution of the Milky Way with Bayesian multilevel models."
2021 May	**Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online** Lightning talk: "Molecular gas in a gravitationally lensed galaxy group at z = 2.9."
2021 Apr	**Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online** "Molecular gas in gravitationally lensed galaxies at z = 2.9."
2020 Oct	**REQUIEM Galaxies Telecon, Online** "Molecular gas in gravitationally lensed galaxies at z = 2.9."

Posters

2021 May	**Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online** "Molecular gas in a gravitationally lensed galaxy group at z = 2.9."
2020 Aug	**Summer Undergraduate Research Program Poster Session, UofT/Online** "Molecular gas in a gravitationally lensed galaxy group at z = 2.9." Awarded prize for one of top three posters.

Outreach

2019 - 2021 **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - end of show.

2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.

Media

2021 Jul **National Ratio Astronomy Organization (NRAO) eNews**
Digital newsletter feature: "Molecular gas in high redshift galaxies". `https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas`

2020 Jul **SURP Student of the Week Feature**
Interview: `https://www.dunlap.utoronto.ca/surp-sow/sow12020/`

Technical Skills

JAX, Stan, Pytorch, SQL, R, C++, Rust, BLAS/LAPACK, LaTeX, regex, bash, git, Docker, Javascript, React